MIABS v0.1: Minimum Information About a Behavioral Simulation

A reporting checklist for published simulations of operant and respondent processes. Every item asks authors to state information rather than to comply with a method. “Not applicable” is always acceptable; silence is not. Draft for public comment | | comments: [OSF project link]

A. Model specification

A1 State every equation or update rule governing the simulated organism, including auxiliary assumptions (response-emission mechanism, choice rule, noise model), not only the focal theoretical equation.

A2 List every free parameter with permissible range, units, and interpretation; identify which are fitted, fixed, or derived.

A3 Distinguish observables (rates, allocations, latencies, IRTs) from latent state variables (associative values, traces, activations); state which quantities predictions are expressed in.

A4 State initial conditions for all state variables and how they were chosen.

B. Environment specification

B1 Specify each contingency completely: schedule type and parameters; interval-timer semantics (arranged vs. obtained; whether timers pause, reset, or run during reinforcement); reinforcer magnitude, duration, and delay; changeover requirements where applicable.

B2 Specify stimulus conditions and their mapping to contingencies, including programmed stimulus–consequence relations in respondent preparations (CS/US durations, ISI, ITI distributions).

B3 State session structure: length or termination criterion, number of sessions, and phase-transition criteria.

C. Temporal structure

C1 State whether time is continuous, discrete, or event-driven; if discrete, state the time step and report sensitivity of results to it.

C2 State how simultaneous events are resolved (e.g., a response coinciding with a timer elapsing) and the order of operations within a step.

D. Stochasticity and reproducibility

D1 Identify every source of randomness in organism and environment, with distributions.

D2 Report random seeds, number of simulated subjects/replicates, and how variability across replicates is summarized.

D3 Provide runnable code and exact dependency versions sufficient to regenerate every figure, or state why not.

E. Parameterization and fitting

E1 State the objective function, fitting algorithm, convergence criteria, and starting-value strategy.

E2 Report evidence that fitted parameters are recoverable: at minimum a parameter-recovery study; where model form permits, a structural identifiability analysis. Report any confounded or unidentifiable parameters.

E3 If models are compared, state the comparison metric and how complexity is accounted for.

F. Validation scope

F1 List the phenomena the simulation is claimed to reproduce, with the qualitative signature of each (i.e., direction, ordering, or functional form, rather than merely “matches the data”).

F2 State known boundary conditions: preparations, schedule ranges, or phenomena where the model is known or expected to fail.

F3 Distinguish results the model was designed or tuned to produce from results that emerged without tuning.

G. Availability

G1 Deposit code, specification files, and generated data in a persistent repository with a DOI; state the license.

Modeled on minimum-information standards in adjacent sciences (MIAME, 2001). Each item is included only because its omission has demonstrably produced a replication failure, hidden confound, or uninterpretable result; the evidence dossier accompanies this document. Version 0.1 | revised annually | [DOI]